PROBLEM Find the value of $(52+6\sqrt{43})^{3/2}-(52-6\sqrt{43})^{3/2}$. **REFERENCE ANSWER** 828

STUDENT Original solution

[37] So $A=(3+\sqrt{43})^2$, and $B=(3-\sqrt{43})^2$

[46] But here, $x=3-\sqrt{43}$, which is negative

[48] $[(3-\sqrt{43})^2]^{3/2}=(3-\sqrt{43})^3$ [...]

[52] $A^{3/2}-B^{3/2}=(3+\sqrt{43})^3-(3-\sqrt{43})^3$

✂ ---------- *cut point*

[68] $(3+\sqrt{43})^3-(3-\sqrt{43})^3=\cdots=140\sqrt{43}$

[69] Answer: $140\sqrt{43}$ ✗

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TEACHER Critique

[48] $[(3-\sqrt{43})^2]^{3/2}=(3-\sqrt{43})^3$

Fix: The expression $(x^2)^{3/2}$ equals $|x|^3$, not x^3. [...] The correct simplification is $\big((3-\sqrt{43})^2\big)^{3/2}=(\sqrt{43}-3)^3$. [...]

[52] $A^{3/2}-B^{3/2}=(3+\sqrt{43})^3-(3-\sqrt{43})^3$

Fix: This step follows the error in [48]. The correct expression is $(3+\sqrt{43})^3-(\sqrt{43}-3)^3$. [...]

The correct final answer is 828.

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STUDENT Rewrite from [52] with the critique

We now compute $(3+\sqrt{43})^3-(\sqrt{43}-3)^3$. Let's expand both cubes.

$(3+\sqrt{43})^3=27+27\sqrt{43}+387+43\sqrt{43}$
$=414+70\sqrt{43}$

$(\sqrt{43}-3)^3=43\sqrt{43}-387+27\sqrt{43}-27$
$=70\sqrt{43}-414$

$(414+70\sqrt{43})-(70\sqrt{43}-414)=828$

Answer: 828 ✓